Introduction

Flight delays are a prevalent problem in airports across the country. Analyzing why and when these delays occur can identify operational bottlenecks to improve airline efficiency. Identifying spatial clustering as well as hot and cold spots of high and low delay will serve as a resource to minimize these delays. Spatial methods allow us to visualize where these delays happen and infer any correlation between delay reasons and among geographically close airports. This project focused on Delta Airlines in particular, analyzing their delay times and causes across the contiguous 48 states. This project sought to answer the following questions: "What are the primary flight delay causes, and do they vary by geographic area?" "Is there any evidence of spatial autocorrelation or clustering in Delta's flight delays?" "How does flight delay time vary by geographical location?"

Data and Methods

The dataset utilized was downloaded from the U.S. Bureau of Transportation Statistics and filtered to include only Delta flights from July 2024 to July 2025. It records all arriving and departing flights for major airlines across the country. The five categories of flight delays in this dataset are carrier, weather, National Air System (NAS), security, and late arriving aircraft. The coordinates for the airports were obtained from a Kaggle dataset and attached using the IATA code. First, I filtered and cleaned the data to include only airports in the contiguous 48 states, and airports for which I had coordinates. I also removed any N/A values, but retained zeros because zero minutes is a valid delay time. The original dataset contained one row of delay data per airport per month. I aggregated these rows to only include one row of delay data per airport by summing the monthly delays. Then I created a column to calculate the primary delay cause, by selecting the "max" for delay cause and assigning a simplified value of either carrier, weather, NAS, security, or late aircraft. I then converted the dataset for preprojection utilizing EPSG 9311 to convert the degrees into meters for mapping.

Next, I used ggplot() to create maps of average flight delay and primary delay cause. I utilized a color scale for both maps, making the map easier to interpret. I then tested for spatial autocorrelation using Moran's I. I converted the data to sp, defined neighbors as the 5 nearest airports, and defined my spatial weights matrix. After running Moran's I, I ran Local Moran's I (LISA) to determine where clustering occurred and identify any hot or cold delay hotspots. I saved the LISA score, p-value, and z-score as columns in my dataset, then used ggplot() to plot the data based on z-score to show significance. I also utilized a color system, red representing hotspots of high delays, blue representing cold spots of low delays, and white representing no clustering of delays. Finally, I created a plot of the delay time by month. I compressed the data to include one row per month by summing delay times per airport. I then graphed this using ggplot(), creating a line graph depicting any trends in seasonality.

Results

The results of the "Average Delay per Flight" map showed that most of the country had an average delay of 10-20 minutes, in the yellow/orange range. There was not initially any visual representation of clusters or spatial autocorrelation, as the delay times seem to be spatially random with an average of approximately 15 minutes. Moving on to the "Primary Delay Cause by Airport" map, the dominant delay cause was carrier-related issues, depicted by pink dots. Late aircraft and NAS delays were the primary causes for some airports, while weather and security were not the primary delay cause anywhere. The Moran's I test resulted in a value of 0.038 with a p-value of 0.171. The Moran's I of nearly 0 confirms that delay levels across U.S. airports appear spatially random, and the high p-value indicates there is no statistically significant clustering of airports with high average delays. The LISA scores remained consistent with these results. Most of the map consisted of very light shades of blue and red, or white. However there was one prominent hot spot in North Dakota, and one cold spot in Minnesota. Lastly, the “Delays by Month” graph revealed the highest delays occurred in July 2024 and the lowest occurred in October 2024. Delay times reach a slight peak again around December and January 2024, then seem to rise again and peak in July 2025, indicating a possible trend.

Conclusions

The findings from this analysis indicate that Delta Airlines’ flight delays across the contiguous United States do not exhibit meaningful spatial clustering. Average delay times were largely consistent nationwide, with most airports experiencing between 10 and 20 minutes of average delay per flight. This pattern appeared visually apparent on the initial delay map and was statistically confirmed through Moran’s I, which returned a value near zero and a non-significant p-value, indicating no global spatial autocorrelation. The Local Moran’s I results further supported this conclusion. Aside from one localized hotspot in North Dakota and a single cold spot in Minnesota, airports did not display significant clustering of unusually high or low delays.

The analysis of primary delay causes revealed that most delay variation stems from carrier-related issues, with late aircraft and NAS delays also contributing at select airports. Weather and security delays did not emerge as dominant causes for any airport in the study period. This conclusion further confirms the earlier findings, as carrier-related issues are specific to an individual aircraft and would theoretically have no effect on another aircraft. Finally, examining delay trends by month showed a clear seasonal pattern, with the highest delays occurring in July 2024 and July 2025, and the lowest delays in October 2024. Modest increases around the winter months suggest that seasonal travel demand may play a role in shaping delay intensity.

Overall, the results suggest that Delta's delay patterns are primarily driven by operational factors rather than geography, and that temporal patterns, particularly those aligned with peak travel seasons, offer more explanatory power than the spatial context.

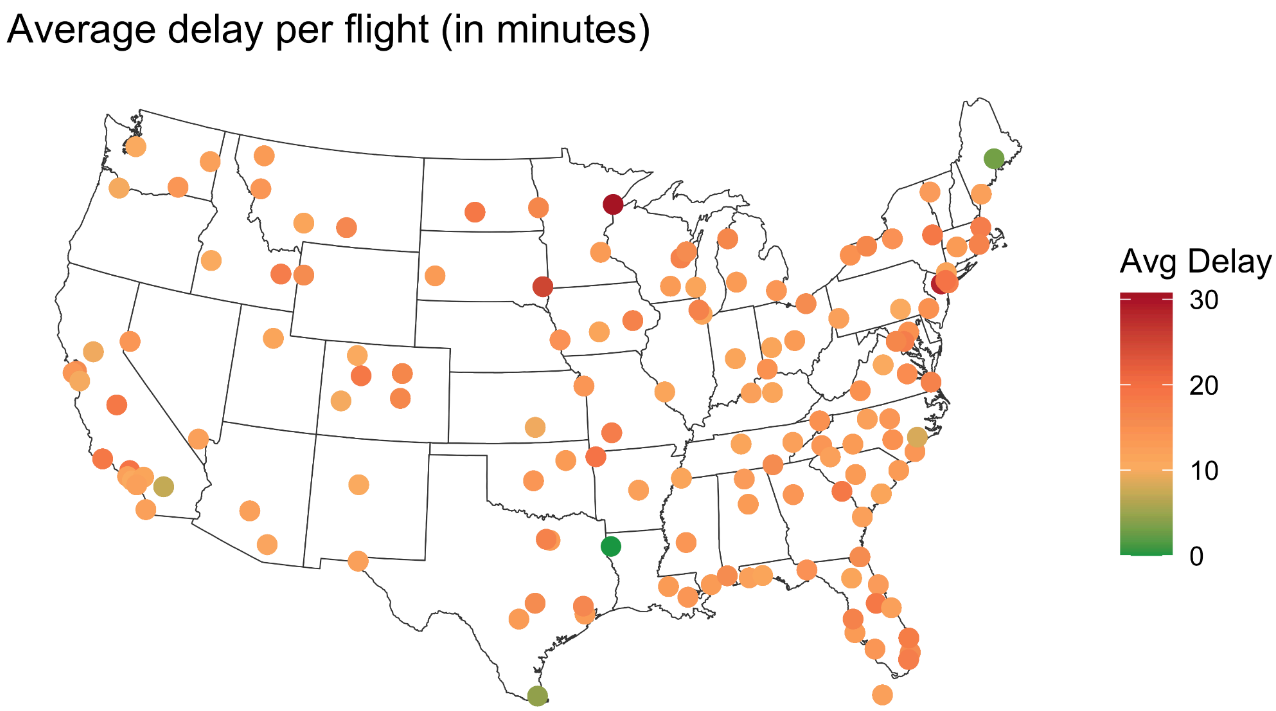
References

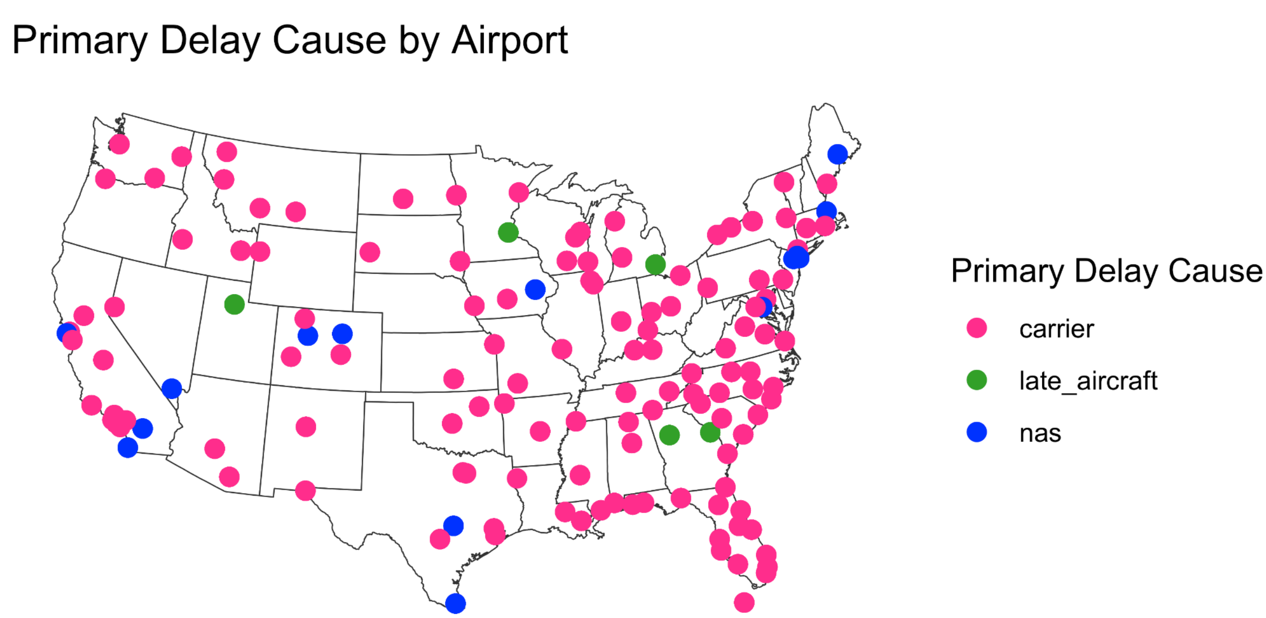
Haroon, S. A. (2023). *Airlines Dataset* [Data set]. Kaggle.

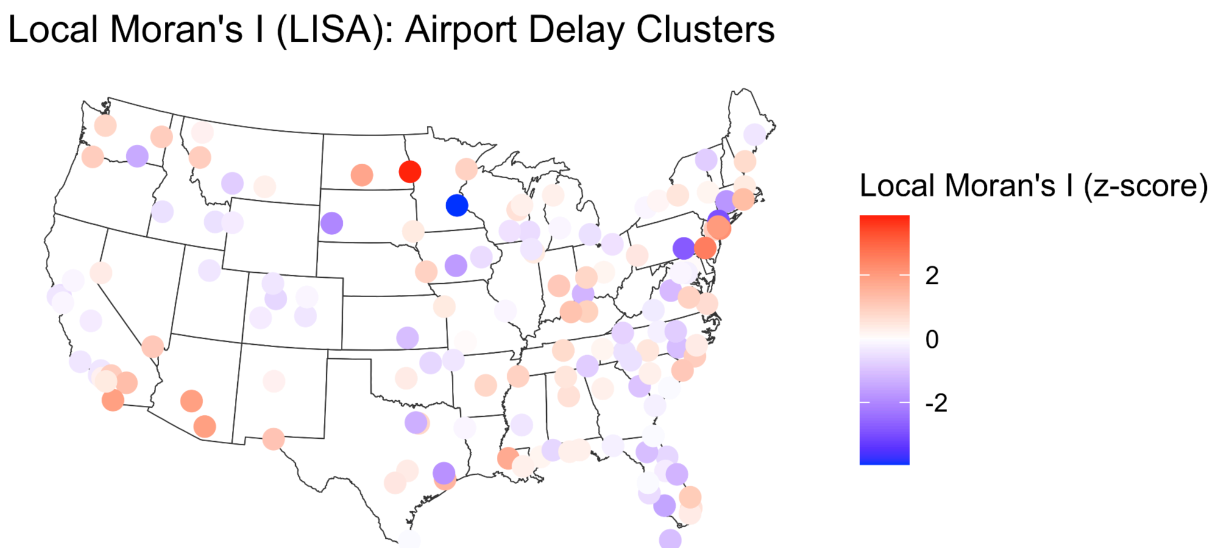
https://www.kaggle.com/datasets/saadharoon27/airlines-dataset

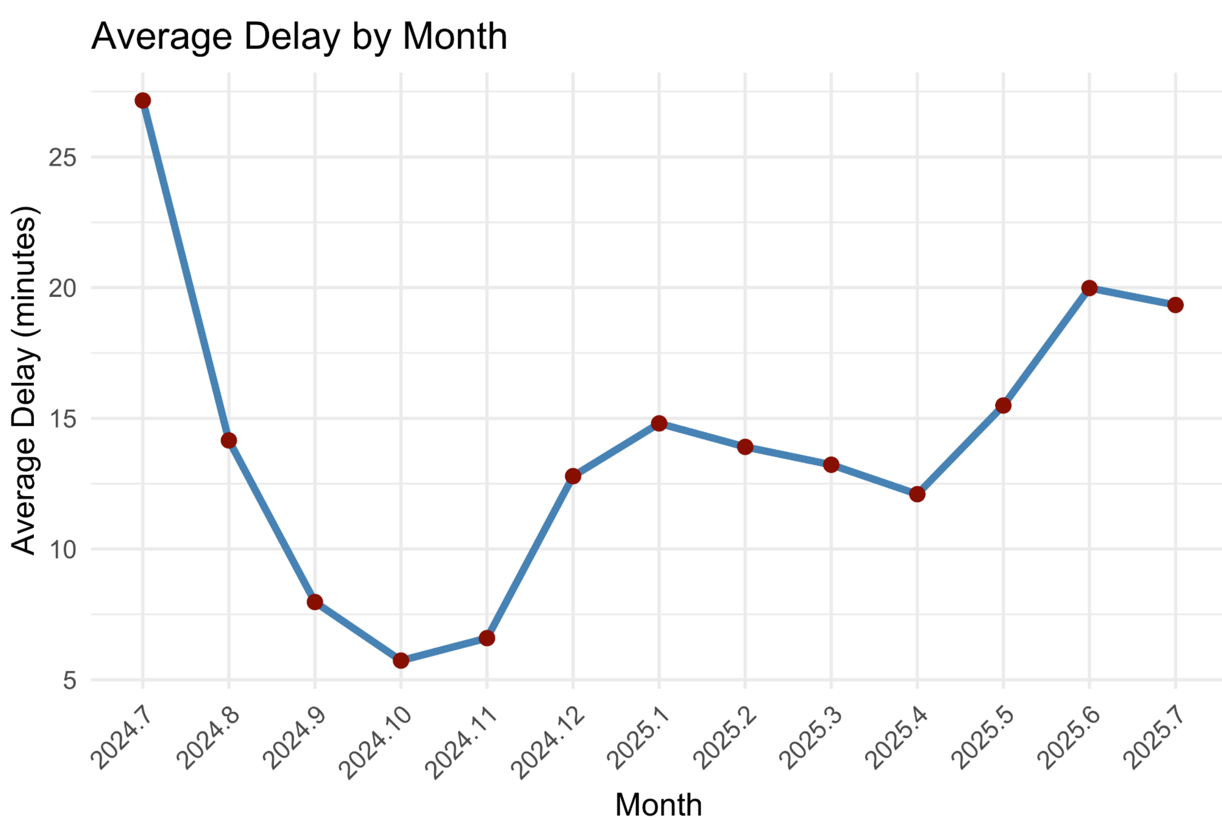
Bureau of Transportation Statistics. (2024). *On-Time Delay Cause Data* [Data set]. U.S. Department of Transportation.

https://www.transtats.bts.gov/ot_delay/OT_DelayCause1.asp?20=E


Average delay per flight (in minutes)
Avg Delay
30
20
10
0


Primary Delay Cause by Airport
Primary Delay Cause
carrier
late_aircraft
nas


Local Moran's I (LISA): Airport Delay Clusters
Local Moran's I (z-score)
2
0
-2


Average Delay by Month
Average Delay (minutes)
25
20
15
10
5
2024.7
2024.8
2024.9
2024.10
2024.11
2024.12
2025.1
2025.2
2025.3
2025.4
2025.5
2025.6
2025.7
Month